

BioDiversity and Environment

A) Choose the best alternatives.

1) We are prohibited to kill endangered species as they have:

- a) ecological value
- b) legal value
- c) economical value
- d) aesthetical value

2) A scientist who studied about genes which are the units in the cells of a living thing that controls its physical characteristics.

- a) an ecologist
- b) a taxonomist
- c) an evolutionist
- d) a geneticist

3) The term 'biodiversity' was coined as a term by

- a) E.O. Wilson
- b) Callicott et al
- c) Roebuck and Phifer
- d) Norton

4) The major cause of degradation biodiversity is.

- a) ~~pollution~~
- b) over population
- c) over consumption
- d) all of them

5) Which one helps to control measures of biodiversity.

- a) pollution
- b) over population
- c) over consumption
- d) sustainable living

B) Very short answer type questions.

1) What is biodiversity?

→ Biodiversity is the variety of living organisms on Earth, including species, genes, and ecosystem.

2) Name the three types of biodiversity.

→ (i) Ecosystem Diversity.

(ii) Species Diversity.

(iii) Genetic Diversity.

3) Write the importance of genetic biodiversity.

→ Genetic biodiversity helps organisms to adapt to environmental changes and increases their ability to survive diseases and other threats.

4) Give two examples of biodiversity.

→ (i) Different varieties of rice are examples of genetic diversity.

(ii) Different species of plants and animals in a forest are examples of species diversity.

5) When and where was the issue of biodiversity brought up by the United Nations?

→ The issue of biodiversity was brought up internationally at the United Nations Conference on Environment and Development (Earth Summit) in 1992 at Rio de Janeiro, Brazil.

6) Who coined the term biodiversity and when?

→ The term biodiversity was coined by Walter G. Rosen in 1985.

7) Define the term 'Species'.

→ A species is a group of similar organisms that can interbreed with one another and produce fertile offspring.

8) Define ecosystem diversity.

→ Ecosystem diversity is the variety of different ecosystems found in a particular area or on the Earth, such as forests, grasslands, rivers and ponds.

9) What is 'species diversity'?

→ Species diversity is the variety of different species of organisms found in a particular area.

10) Short answer the questions.

1) Write the causes of loss of biodiversity.

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- Deforestation and habitat destruction.
 - Overpopulation
 - Overexploitation of natural resources
 - Pollution
 - Climate change
 - Hunting and poaching
 - Introduction of invasive species

3) Why is it important to attach aesthetic value to nature?

→ Nature has beauty and provides enjoyment, peace and recreation. Appreciating its aesthetic value encourages people to love, respect and conserve nature and biodiversity.

2) Why is it important to attach obtain knowledge of biodiversity?

- Knowledge of biodiversity helps us to:
- Understand different living organisms and their relations.
 - Identify and conserve endangered species.
 - Use natural resources wisely.
 - Maintain ecological balance.

4) Write the importance of biodiversity.

- Biodiversity is important because it:
- Maintains ecological balance.
 - Provides food, medicine and raw materials.
 - Supports ecosystem services.
 - Provides genetic resources.
 - Supports human life and economic activities.

5) Write the methods of conservation of biodiversity.

→ The two main methods of ~~con~~ biodiversity conservation are:-

- (i) In-situ conservation: Conservation of organisms in their natural habitats such as national parks and conservation areas.
- (ii) Ex-situ conservation: Conservation of organisms outside their natural habitats, such as zoos, botanical gardens and gene banks.

6) Write the benefits of biodiversity.

→ The benefits of biodiversity are:-

- It provides food and medicine.
- It supplies raw materials.
- It maintains environmental balance.
- It provides genetic resources.
- It supports agriculture and other human activities.

7) What is the role of biodiversity in maintaining environmental balance?

→ Biodiversity maintains environmental balance by maintaining food chains and food webs, nutrient cycles and ecosystem functions. Different organisms depend on one another so biodiversity helps keep ecosystems stable.

8) Write the differences between species biodiversity and genetic biodiversity.

Species Biodiversity	Genetic Biodiversity
(i) It is the variety of different species in an area.	(i) It is the variation among individuals of the same species.
(ii) Examples: Different species of birds in a forest.	(ii) Examples: Different varieties of rice.

9) What is the human impact on biodiversity?

→ Human activities such as deforestation, urbanization, pollution, overhunting, overgrazing and excessive use of natural resources destroy habitats and reduce biodiversity. They may also cause extinction of species.

10) What are the various measures to conserve biodiversity?

→ The major measures are:

- Protecting natural habitats and forests.
- Establishing national parks and conservation areas.
- Controlling hunting and poaching.
- Controlling pollution.
- Practising afforestation and reforestation.
- Using natural resources sustainably.
- Conserving endangered species.
- Creating public awareness.

D) Long answer questions:

1) How are valuation studies of biodiversity useful?

→ Valuation studies help us understand the economic, social, cultural and environmental value of biodiversity.

* They are useful because:

(1) They show the importance of biodiversity to human beings.

(2) They help in making plans and policies for conservation.

(3) They help determine the economic value of natural resources.

(4) They encourage people to conserve biodiversity.

(5) They help governments decide how natural resources should be used sustainably.

Thus, valuation studies help in the proper use and conservation of biodiversity.

2) How did traditional societies preserve the earth's biodiversity?

→ Traditional societies preserved biodiversity through their close relationship with nature. They:

(1) Used natural resources according to their needs.

(2) Avoided unnecessary destruction of forests.

(3) Protected sacred forests, ponds and other natural areas.

(4) Followed traditional rules against hunting and overexploitation.

(5) Conserved local varieties of plants and animals.

(6) Practised traditional farming methods.

(7) Passed knowledge about nature and conservation from one generation to another.

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Thus, traditional practices helped maintain biodiversity and ecological balance.

- 3) How is genetic diversity expressed?
- Genetic diversity is expressed through differences in the genetic characteristics of individuals of the same species. These differences may be seen in size, colour, shape, height, resistance to diseases and other characteristics.

For example, humans belong to the same species, but they have differences in skin colour, hair type, eye colour, height and body structure. Similarly, different varieties of rice, wheat and maize show genetic diversity.

- Q) How can we conserve biodiversity for human substance?

- We can conserve biodiversity for human substance ~~are~~ by:-
- (i) Protecting forest.
 - (ii) Protecting natural habitats.
 - (iii) Preventing illegal hunting and poaching.
 - (iv) Establishing natural parks, wildlife reserves and conservation areas.
 - (v) Conserving endangered plants and animals.
 - (vi) Controlling pollution.
 - (vii) Using natural resources wisely etc.

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5) Biodiversity also has a role to play in the stability of the ecosystem and global climate change. Explain with justification.

→ Biodiversity helps maintain the stability of the ecosystem and reduces the effects of climate change.

(i) Different organisms maintain food chains and food webs.

(ii) Plants absorb carbon dioxide (CO_2) and release oxygen.

(iii) Forests store carbon and help reduce global warming.

(iv) Organisms help in decomposition and nutrient cycling.

(v) Forests and wetlands help regulate water temperature and climate.

Therefore, biodiversity is essential for a stable ecosystem and helps reduce the harmful effects of global climate change.

6) "The greatest threat leading to the loss of biodiversity is the human race". Justify the statement.

→ Human activities are a major cause of biodiversity loss because they destroy natural habitats and harm living organisms.

(i) Deforestation destroys forests and habitats.

(ii) Urbanization destroys natural areas.

(iii) Overhunting and poaching reduce wildlife populations.

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- (iv) Pollution harms plants and animals.
 - (v) Overexploitation reduces natural resources.
 - (vi) Overgrazing damages vegetation and soil.
 - (vii) Climate change affects ecosystem and species.
 - (viii) Invasive species compete with native species.

Therefore, human activities are a major threat to biodiversity, so we must conserve natural resources and protect the environment.

7) Could the loss of biodiversity be a greater threat to humanity than climate change? Justify.

→ Yes, the loss of biodiversity could be a greater threat to humanity because:

- (i) It can cause the extinction of plants and animals.
- (ii) It reduces our sources of food, medicine and raw materials.
- (iii) It disturbs food chains and ecosystem balance.
- (iv) It can reduce pollination, soil fertility and clean water.
- (v) It makes ecosystems less ~~is a serious~~ able to cope with climate change and natural disasters.

• Therefore biodiversity loss is a serious threat to human survival and must be conserved along with controlling climate change.

8) The diagram given below shows the one of the methods of conservation of biodiversity. study the diagram answer the following question.

(a) Mention the type of control measures of biodiversity.

→ The type of control measures of biodiversity is In-situ conservation.

b) Write its advantages.

→ Its advantage ~~are~~ is to protect organisms ~~in~~ in their natural habitat and maintains the natural ecosystem.

c) How does it help in the conservation of biodiversity?

→ It protects plants and animals along with their natural habitats allowing them to ~~pr~~ reproduce and maintain biodiversity naturally.

9) study the diagram and answer the following questions.

(a) Mention the type of method of biodiversity conservation.

→ In-situ Conservation (National Park)

b) Write its any two advantages.

→ (i) It protects endangered and ~~rare~~ rare species in their natural habitats.

(ii) It protects their habitats and maintains ecological balance.

(c) How does it help in the biodiversity conservation?
→ It provides a protected natural area where plants and animals can live, reproduce and survive without harmful human activities.

16) Seeing a crowd of students in one corner of the school, the principal rushed to see the matter and found some children beating and chasing a ~~papa~~ frog. On seeing the principal all the children fled to their classes except Dinesh who requested the principal to arrange for some medical assistance for the injured animal. The principal rewarded him for his attitude.

a) Species of few animals are very less in number. What are these species known as?

→ Endangered species.

b) Mention the threat to biodiversity.

→ Hunting and poaching of animals are threats to biodiversity.

c) It is wrong to harm the animals. Give reason.

→ Animals are an important part of the ecosystem. Harming or killing them disturbs ecological balance and may cause their extinction.

d) Give example of some endemic species of Nepal.

→ Examples include spring babbler, Himalayan field mouse and Himalayan newt.

* Unit 4A *
* The End *

Sun
12/3/2018